Data and Metadata Management: A Business Perspective

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Credits

- Slides created by Wendy Thomas and Marcel Hebing
- Slides footnoted with a deck number (i.e., S11):
  - The slides were developed for several DDI workshops at IASSIST conferences and at GESIS training in Dagstuhl/Germany
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  - Further contributors
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    - Pascal Heus, Open Data Foundation
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PART I

Introduction to DDI

Introduction to DDI

WELCOME AND OUTLINE

Outline

Introduction to DDI

- Welcome
- DDI in 60 Seconds
- Processes
- DDI-Standard
- Tools
- Codebook and Lifecycle

Advanced Topics

- DDI-L in Detail
- Schemes and Reuse
- Building a codebook
- Capturing a process
  - Questionnaire
- Discovery & Comparison
- Questions & Conclusion

Introductions

- Who are you?
- What does your organization do?
  - Data collection
  - Data production
  - User access
  - Preservation
- What is the scale of your operations?

Changes in the environment:

- Expanded access to data
- Data available through multiple portals
- Cross portal access
- Linking and layering of data from different sources and different disciplines
- Cost of developing system specific software
- Cost of non-interoperability over the life of the data

The Challenge

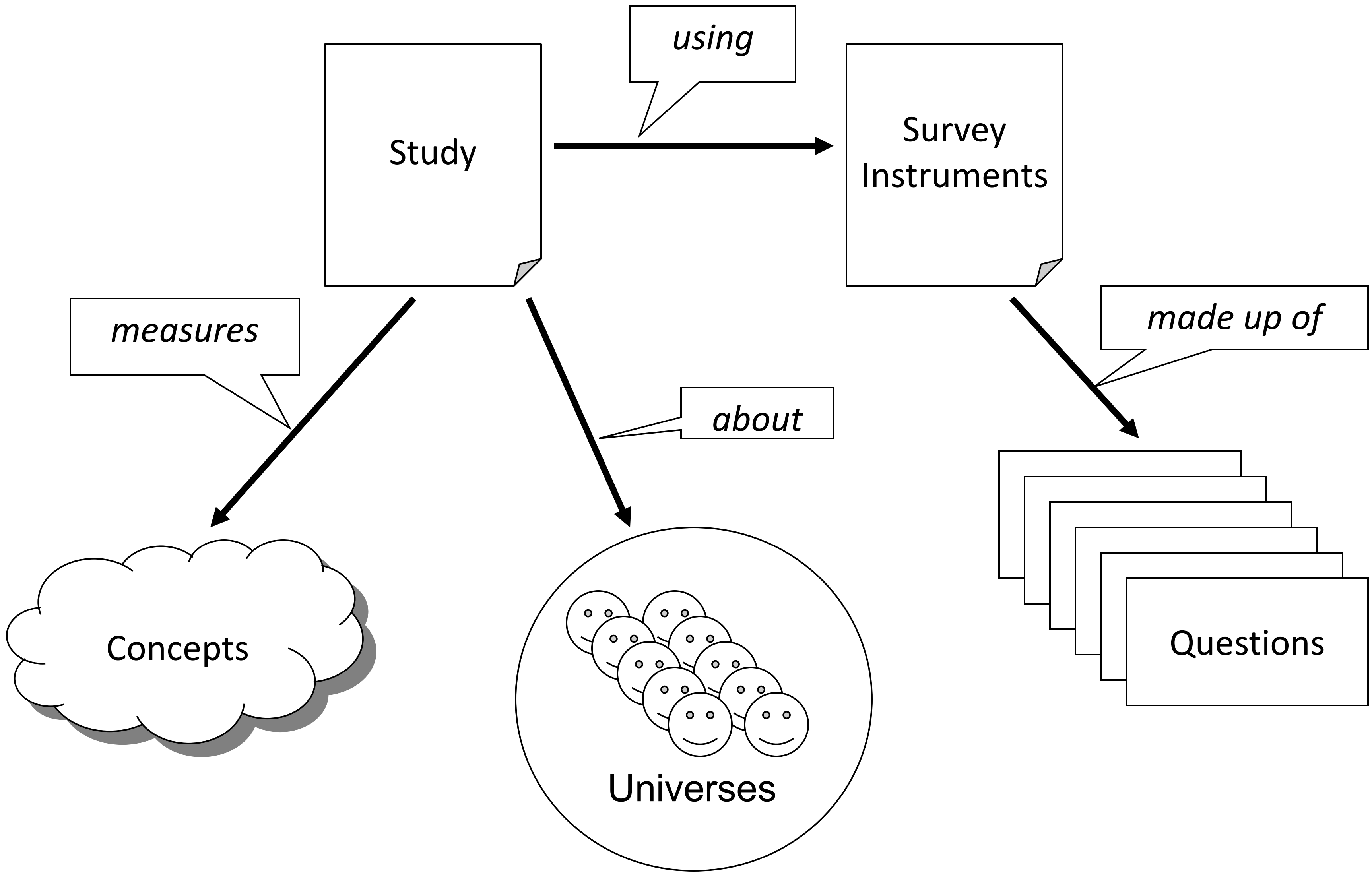
- In a large organization, there are many different streams of data production
- “Silos” tend to emerge, each with a different set of systems and processes
- This makes the management of data, metadata, and the production process very difficult – the different systems/silos don’t interoperate easily
- It is difficult to realize the vision of “industrialized” data production, with its attendant efficiencies

Why Such a Mess?

- Everyone thinks their data are special
- And they are right – they are special…
- They're just not as special as they think!
- The point is that good IT solutions for data management can be built on the basis of the *similarities* across the silos
  - Computers can't think!
  - They manipulate the structural aspects of data
  - The structural aspects of data are the same

Introduction to DDI

DDI 3 IN 60 SECONDS



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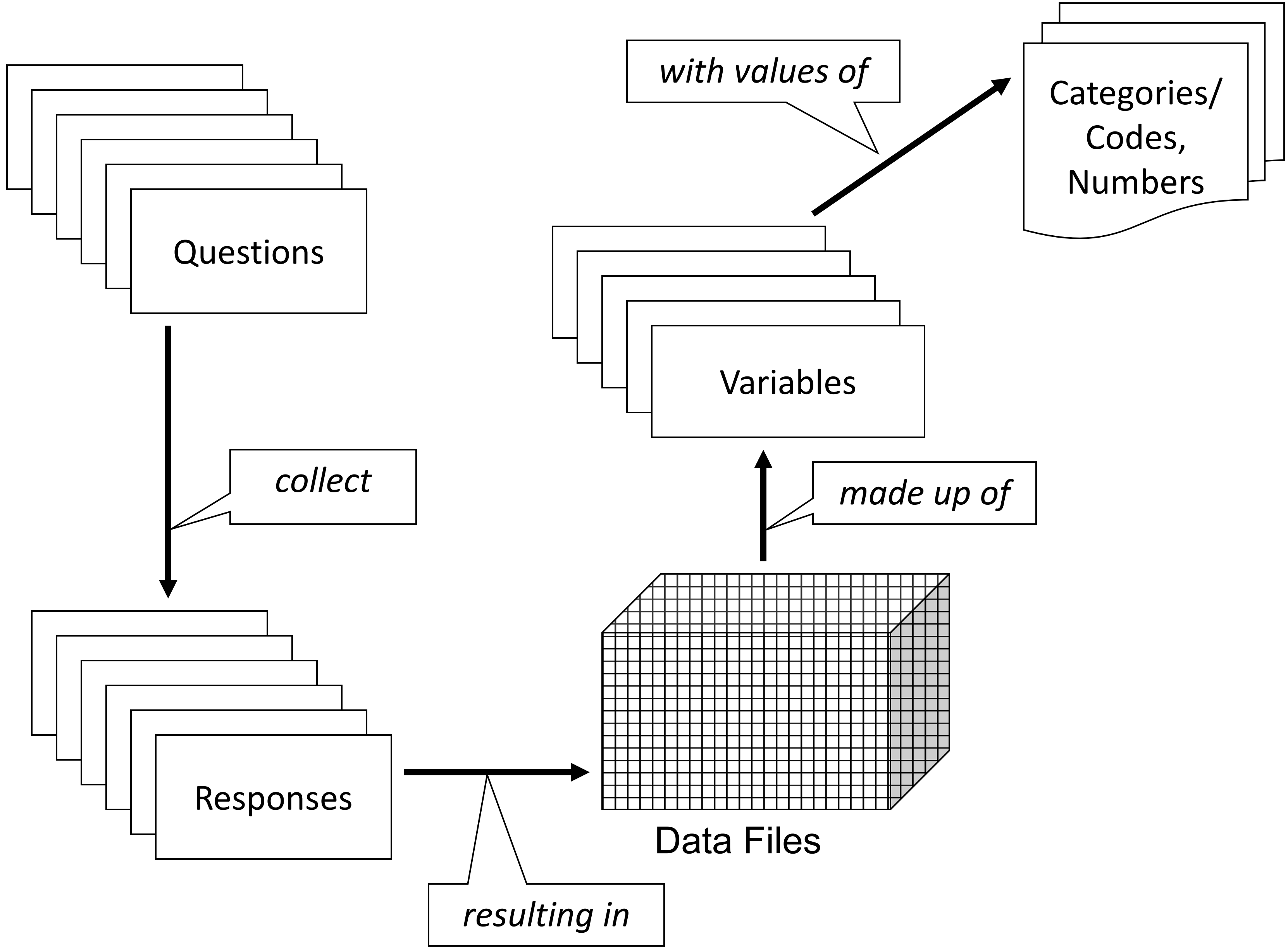

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Introduction to DDI

PROCESSES

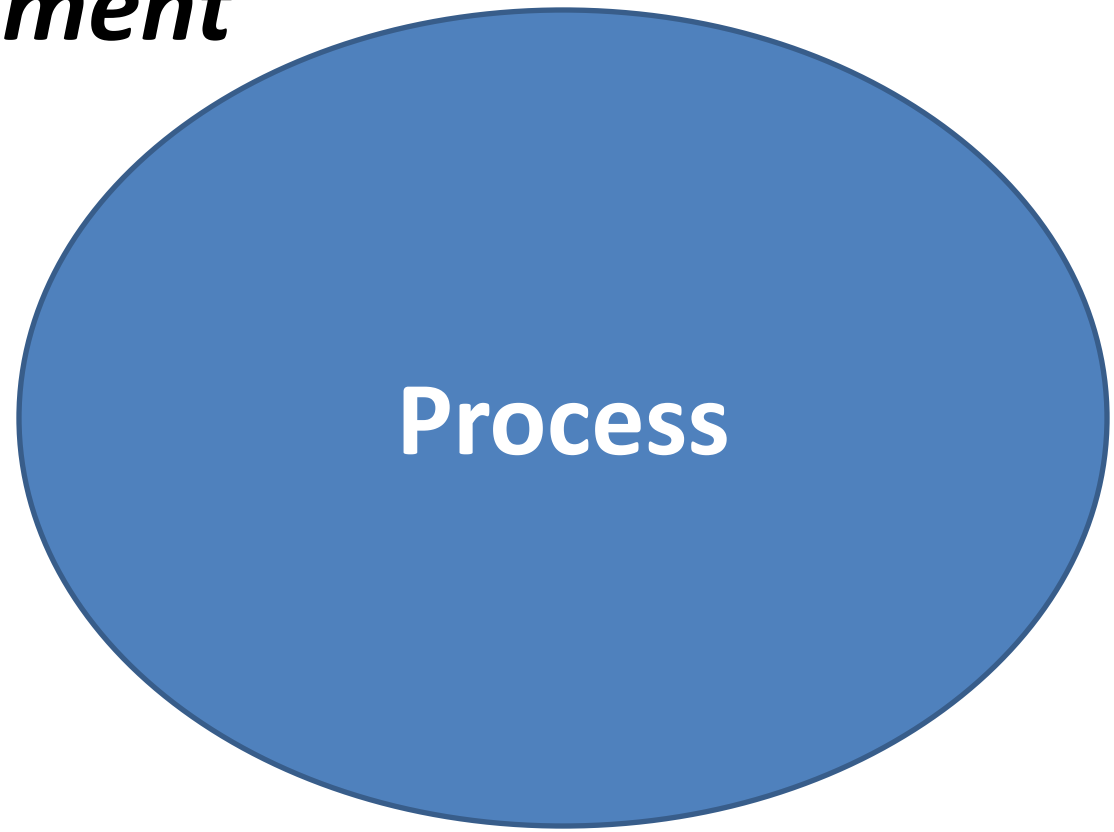
Looking at your organization

- What activities take place and what materials do they involve?
- What specific processes take place and in what order?
- Which processes produce metadata of what type?
- What are the critical activities or processes?
- What do you control?

Traditional Process Model

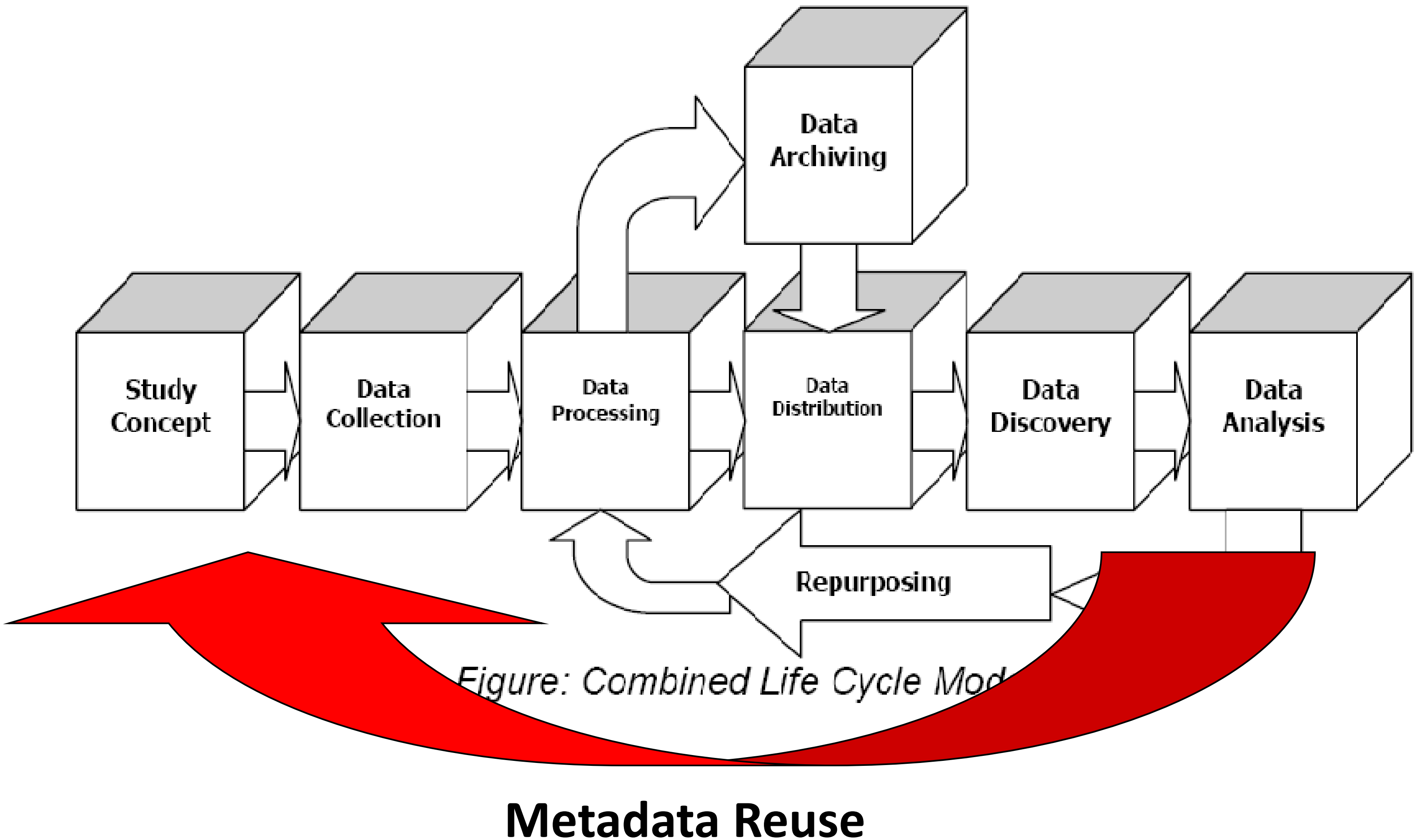
Environment

INPUT



OUTPUT

DDI Lifecycle Model



Quality Management / Metadata Management

1 Specify Needs	2 Design	3 Build	4 Collect	5 Process	6 Analyse	7 Disseminate	8 Archive	9 Evaluate
1.1 Determine needs for information	2.1 Design outputs	3.1 Build data collection instrument	4.1 Select sample	5.1 Integrate data	6.1 Prepare draft outputs	7.1 Update output systems	8.1 Define archive rules	9.1 Gather evaluation inputs
1.2 Consult & confirm needs	2.2 Design variable descriptions	3.2 Build or enhance process components	4.2 Set up collection	5.2 Classify & code	6.2 Validate outputs	7.2 Produce dissemination products	8.2 Manage archive repository	9.2 Conduct evaluation
1.3 Establish output objectives	2.3 Design data collection methodology	3.3 Configure workflows	4.3 Run collection	5.3 Review, Validate & edit	6.3 Scrutinize & explain	7.3 Manage release of dissemination products	8.3 Preserve data and associated metadata	9.3 Agree action plan
1.4 Identify concepts	2.4 Design frame & sample methodology	3.4 Test production system	4.4 Finalize collection	5.4 Impute	6.4 Apply disclosure control	7.4 Promote dissemination products	8.4 Dispose of data & associated metadata	
1.5 Check data availability	2.5 Design statistical processing methodology	3.5 Test statistical business process		5.5 Derive new variables & statistical units	6.5 Finalize outputs	7.5 Manage user support		
1.6 Prepare business case	2.6 Design production systems & workflow	3.6 Finalize production system		5.6 Calculate weights				
				5.7 Calculate aggregates				
				5.8 Finalize data files				

Generic Longitudinal Business Process Model: Overview

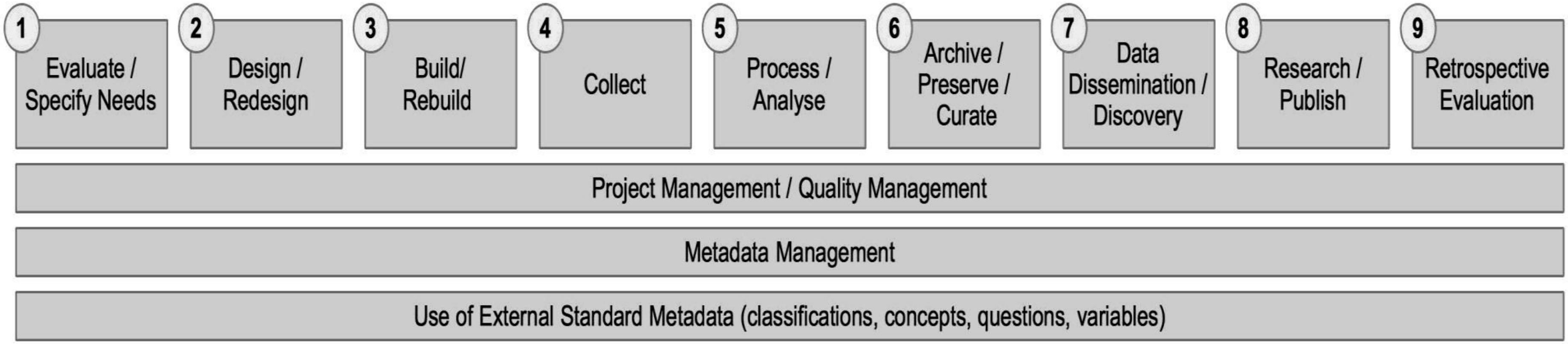
1 Evaluate / Specify Needs	2 Design / Redesign	3 Build / Rebuild	4 Collect	5 Process / Analyse	6 Archive / Preserve / Curate	7 Data / Dissemination / Discovery	8 Research / Publish	9 Retrospective Evaluation
1.1 Define research questions, universe & high-level concepts	2.1 Identify sources	3.1 Develop data collection instruments	4.1 Select sample	5.1 Integrate data	6.1 Ingest data & metadata	7.1 Deploy release infrastructure	8.1 Discovery of data and relevant research	9.1 Establish evaluation criteria
1.2 Evaluate existing data & publications	2.2 Design sampling methods	3.2 Create or enhance infrastructure components	4.2 Set up collection	5.2 Classify & code	6.2 Enhance metadata	7.2 Preserve dissemination products	8.2 Access data	9.2 Gather evaluation inputs
1.3 Establish outputs & needed infrastructure	2.3 Design collection instruments	3.3 Validate instruments	4.3 Run collection	5.3 Explore, validate & clean data	6.3 Preserve data & metadata	7.3 Deploy access control system / policies	8.3 Prepare data	9.3 Conduct evaluation
1.4 Define specific concepts to be measured	2.4 Specify data elements	3.4 Test production systems	4.4 Finalize collection	5.4 Impute missing data	6.4 Ongoing Curation	7.4 Promote dissemination products	8.4 Analyze data	9.4 Determine future actions
1.5 Planning, timetable & needed infrastructure	2.5 Specify processing / data cleaning methods	3.5 Finalize production systems		5.5 Construct new variables and units		7.5 Provide data citation support	8.5 Prepare research papers	
1.6 Prepare proposal and get fundng	2.6 Specify analysis plan			5.6 Calculate weights		7.6 Enhance data discovery	8.6 Manage disclosure risk	
	2.7 Organize research team			5.7 Calculate aggregates		7.7 Manage user support	8.7 Publish research	
	2.8 Design infrastructure			5.8 Anonymize data				
				5.9 Finalize data outputs				

Project Management / Quality Management

Metadata Management

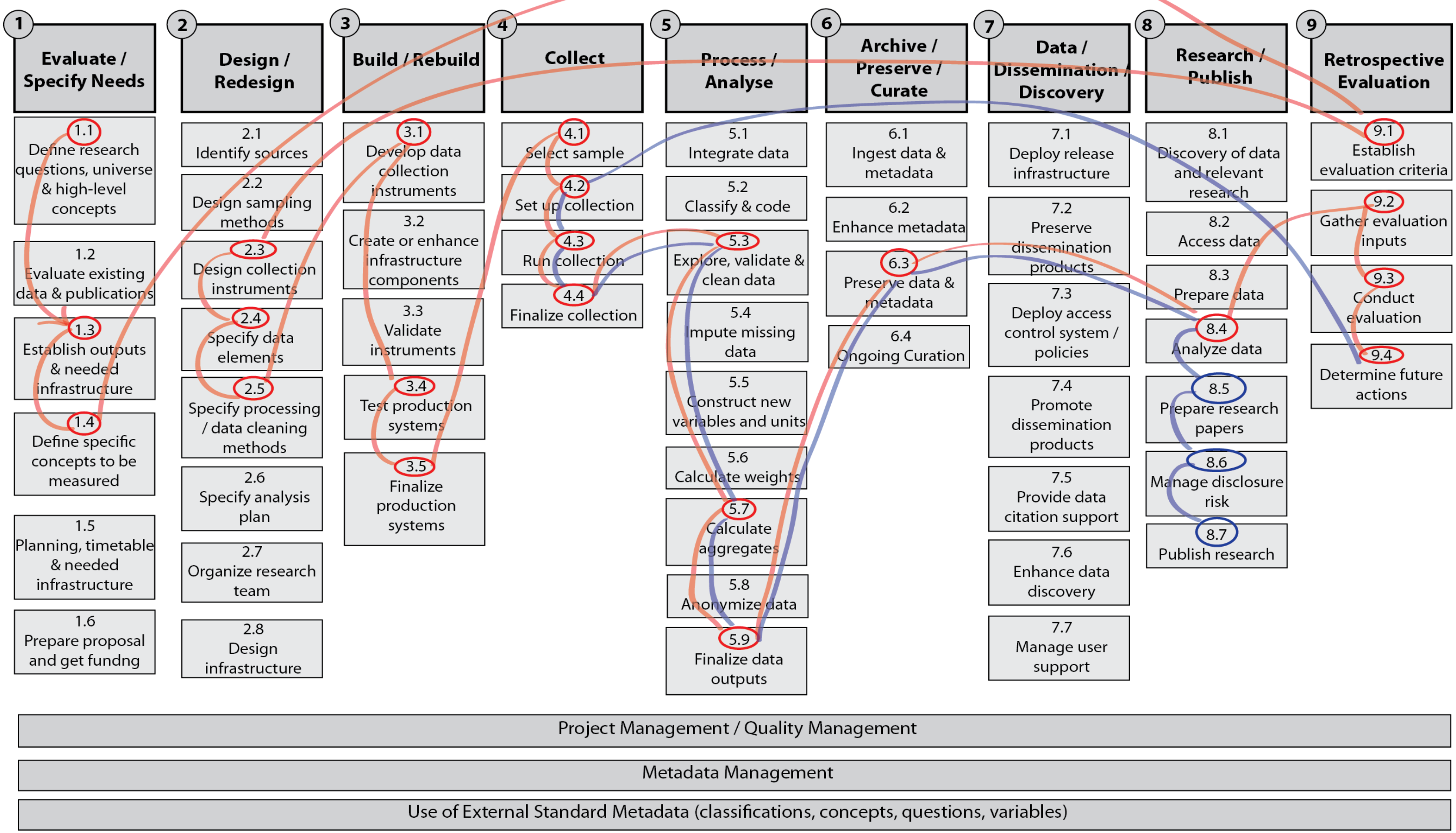
Use of External Standard Metadata (classifications, concepts, questions, variables)

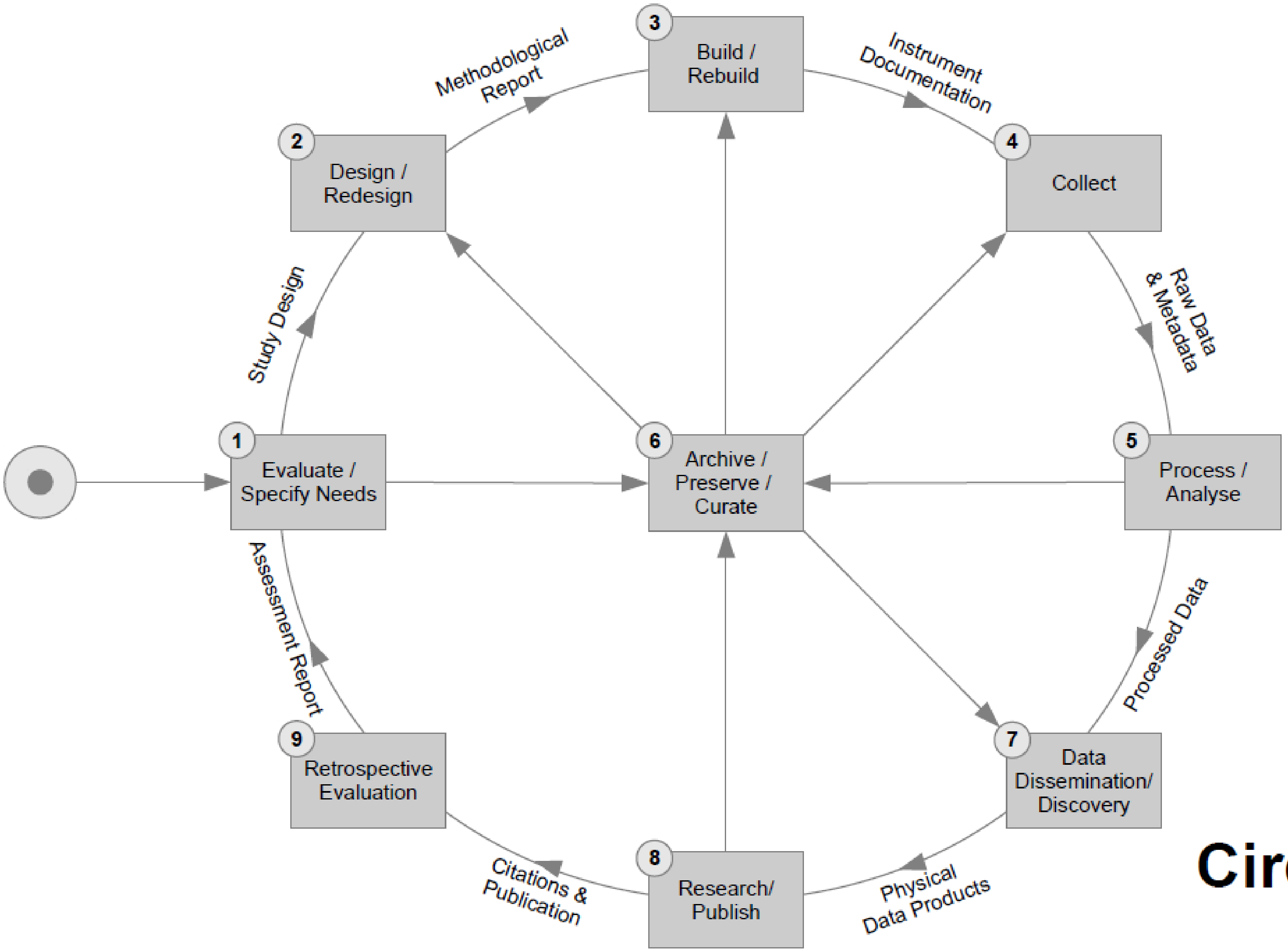
Generic Longitudinal Business Process Model: **High Level View**



Note the similarity to the DDI Combined Lifecycle Model and the top level of the GSBPM

Generic Longitudinal Business Process Model: Repeated Survey, One Panel, No Redesign, One Publication
First round is Red, Second Round is Blue. Starts at 1.1, ends at 8.7





3
Build /
Rebuild
Methodological
Report
Instrument
Documentation
2
Design /
Redesign
4
Collect
Study Design
Raw Data
& Metadata
1
Evaluate /
Specify Needs
6
Archive /
Preserve /
Curate
5
Process /
Analyse
Assessment Report
Processed Data
9
Retrospective
Evaluation
7
Data
Dissemination/
Discovery
8
Research/
Publish
Citations &
Publication
Physical
Data Products

Circle View

Introduction to DDI

THE DDI-STANDARD

What Is DDI?

- An international specification for structured metadata describing social, behavioral, and economic data
- A standardized framework to maintain and exchange documentation/metadata
- DDI metadata accompanies and enables data conceptualization, collection, processing, distribution, discovery, analysis, repurposing, and archiving.
- A basis on which to build software tools
- Currently expressed in XML – e**X**tensible **M**arkup **L**anguage

History

- 1995 -- First international committee established
- 2000 -- First DDI version published (aligned with codebooks, XML DTD-based)
- 2003 – DDI 2 published (support for aggregate/tabular data and geography added)
- 2003 -- Formation of the DDI Alliance, a self-sustaining membership organization
- 2008 – DDI 3 published (aligned with data lifecycle, XML Schema-based)
- 2010 – DDI rebranding – DDI Codebook (DDI 2 branch) and DDI Lifecycle (DDI 3 branch) development lines
- Today – DDI Moving Forward

XML Elements

<Book>

<Title> ***The Hitchhiker's Guide to the Galaxy*** </Title>

<Author> Douglas Adams </Author>

<Year> 1979 </Year>

</Book>

XML Attributes

<Book language=“English”>

<Title> ***The Hitchhiker's Guide to the Galaxy*** </Title>

<Author> Douglas Adams </Author>

<Year> 1979 </Year>

</Book>

Conflicting Tag Names

```xml
<MyData>
	<Table>
		<Legs>4</Legs>
		<Length units="feet">5</Length>
		<Width units="feet">3</Width>
	</Table>

	<Table>
		<Rows>4</Rows>
		<Columns>3</Columns>
	</Table>
</MyData>
```

```xml
<MyData
      xmlns:kitchen="http://www.example.org/kitchen"
      xmlns:data="http://www.example.org/data">

      <kitchen:Table>
            <Legs>4</Legs>
            <Length units="feet">5</Length>
            <Width units="feet">3</Width>
      </kitchen:Table>

      <data:Table>
            <Rows>4</Rows>
            <Columns>3</Columns>
      </data:Table>
</MyData>
```

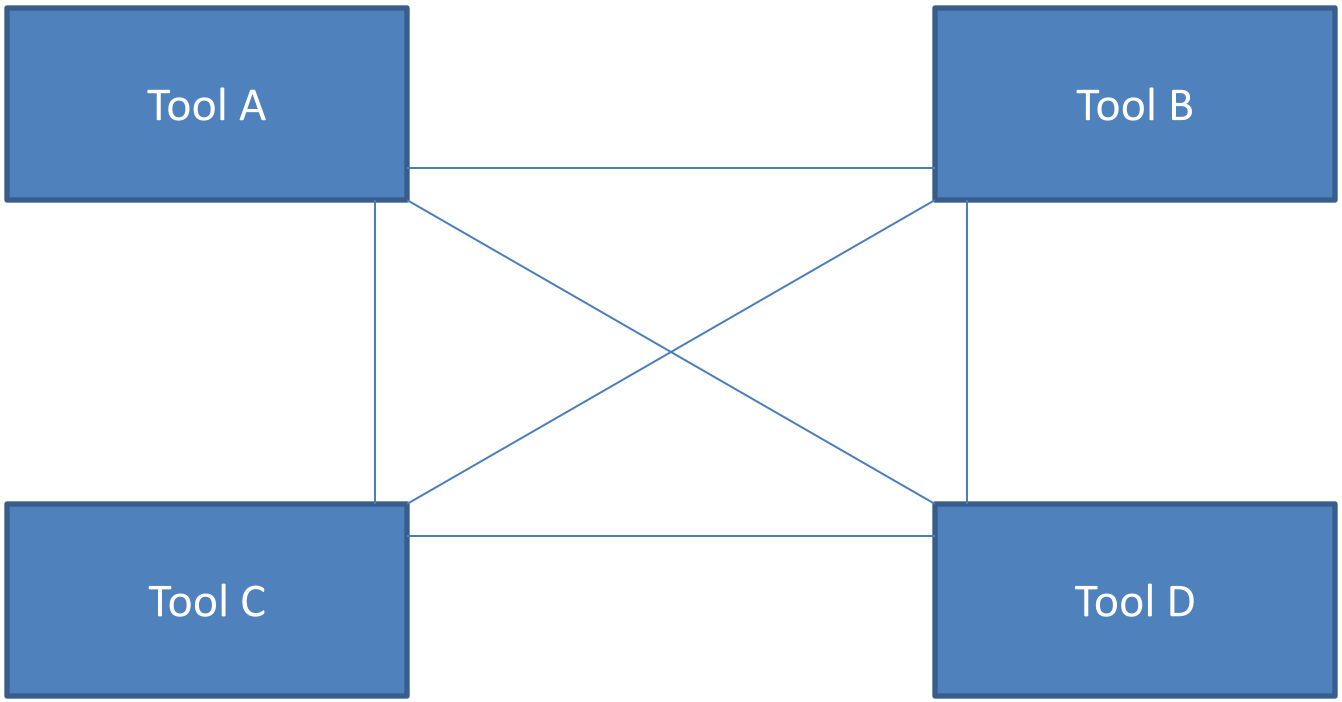
DDI and XML

```
<DDIInstance>
    <StudyUnit> ... </StudyUnit>

    <ResourcePackage>
        <QuestionScheme>...</QuestionScheme>
        <VariableScheme>...</VariableScheme>
        <ConceptScheme>...</ConceptScheme>
        <PhysicalInstance>...</PhysicalInstance>
    </ResourcePackage>

</DDIInstance>
```

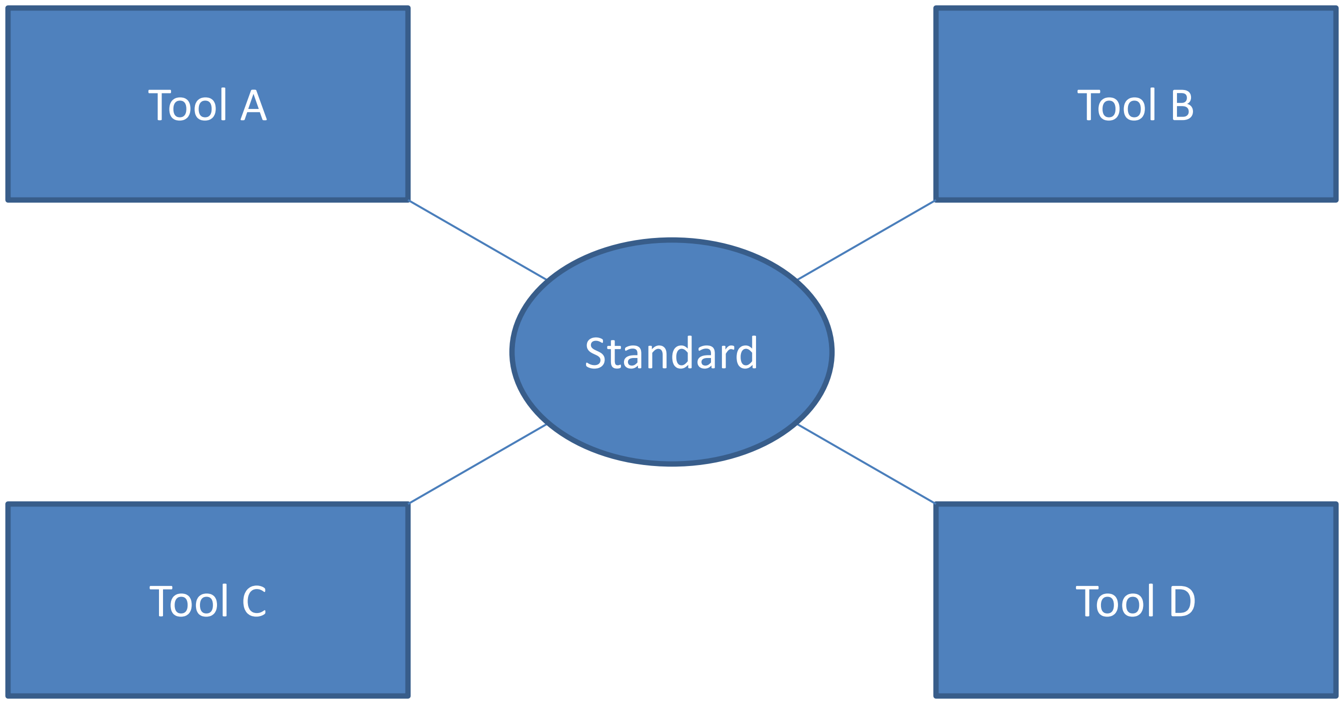
Introduction to DDI

TOOLS

Tools and Standards



Tools and Standards



Examples

- Missy
- IHSN (NADA)
- Questasy
- Nesstar
- Colectica
- Stat/Transfer
- DDI on Rails

Introduction to DDI

CODEBOOK AND LIFECYCLE … AND MOVING FORWARD

Formerly known as…

- DDI 2 → DDI Codebook → DDI-C
- DDI 3 → DDI Lifecycle → DDI-L

DDI Codebook

Parts:

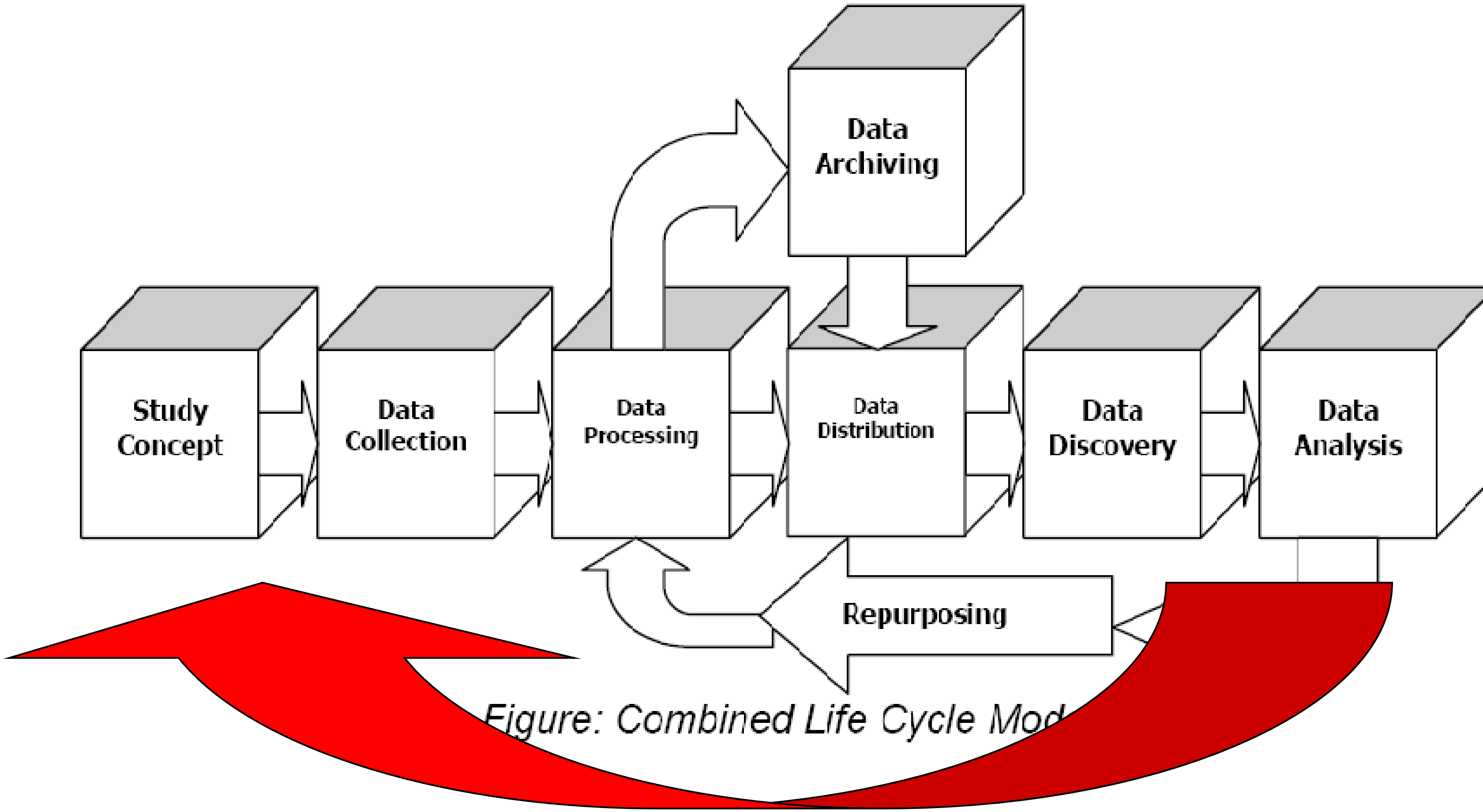
- Document Description
- Study Description
- File Description
- Variable Description

DDI Codebook

From a developer`s perspective:

- Clear design
- Easy to implement
- Interoperability is realistic

DDI Lifecycle Model



Metadata Reuse

DDI Lifecycle Features

- Machine-actionable
- Modular and extensible
- Multi-lingual
- Aligned with other metadata standards
- Can carry data in-line
- Focused on metadata reuse

DDI Lifecycle Features

- Support for CAI instruments
- Support for longitudinal surveys
- Focus on comparison, both by design and after-the-fact (harmonization)
- Robust record and file linkages for complex data files
- Support for geographic content (shape and boundary files)
- Capability for registries and question banks

DDI Codebook

Easy to implement

DDI Lifecycle

Framework

DDI Moving Forward

Model based design

PART II

Advanced Topics

Advanced Topics

DDI LIFECYCLE IN DETAIL

DDI Objective: Organizing Information

- Capture information at point of origin
- Manage information of different types
- Tie together related pieces of information and publish a coherent set of information about a data set, study, collection, or activity
- Understand how information from different data sets relate to each other
- Discover current and future relationships between data and metadata sources

Why am I doing this?

- I need to manage my processes/activities
  - Quality control
  - Replicate processes
  - Tracking provenance
- I need to help analysts locate and use my data accurately (What, who, why, and how of the data)
- I need to interact with other statistical organizations; share metadata, data, and tools

Advanced Topics

SCHEMES AND REUSE

DDI Schemes and Modules

- DDI has two important structures:
  - “Schemes”
  - “Modules”
- A scheme is a list of reusable metadata items of a specific type
- A module is a package of metadata corresponding to a stage of the lifecycle or a specific structural function

Metadata Captured in Schemas

- Foundational objects
  - Concepts
  - Universe (population)
  - Representations
  - Categories / Code Lists
- Instantiations
  - Variables (logical content)
  - Physical structure
  - Questions
  - Instruments
- Processes
  - Methodology
  - Data capture
  - Data processing
- Management
  - Identification
  - Access / Ownership
  - Organization / Individual
  - Coverage (topical, spatial, temporal)
  - Lifecycle events
  - Quality assessment

Schemes

- Concept
- Universe
- Conceptual Variable
- Represented Variable
- Geographic Structure
- Geographic Location
- Managed Representation
- Category
- Code List
- Variable
- NCube
- Physical Structure
- Record Layout
- Question
- Control Construct (flow)
- Interviewer Instructions
- Instrument
- Processing Events
- Processing Instructions
- Organization
- Quality Statement

Managed Content

- Schemes are designed to manage content over time
- Including full or partial schemes or individual objects in a study by reference has the following advantages:
  - Explicit comparability between multiple studies using the same object
  - Retention of the relationship of a set of objects to a larger structure (levels of a hierarchical classification, subsets of geography, etc.)
  - Retention of the information on how an object or set of objects changes over time

Managed Content (cont.)

- Multiple copies result in multiple variations
- Managed content within a registry can notify downstream users of coming changes
  - Change in a Question Response Domain in an economic survey could effect the aggregated data that is used by another process downstream
  - Early notification of the change allows evaluation and modeling of the impact the change may have on the data

Scheme structure

- Name, label, description
- Inclusion of another scheme by reference
- List of included versionable objects (in-line or by reference)
- Grouping of objects
  - Name, Label, Description, and Type
  - Concept / Universe associated with the group
  - Subject / Keyword classification of the group
  - Inclusion of objects or groups by reference
  - Can be ordered

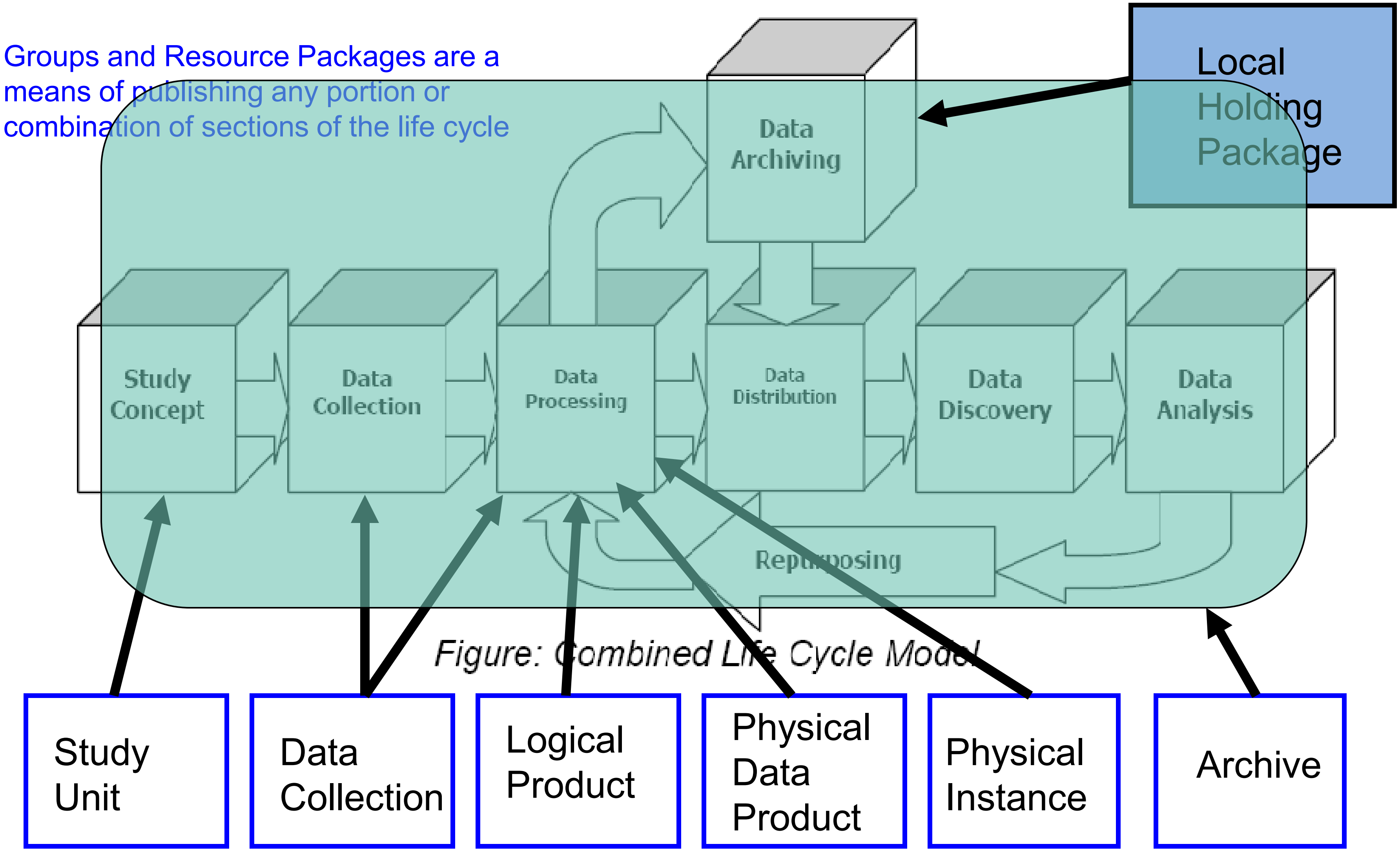
Advanced Topics

BUILDING A CODEBOOK

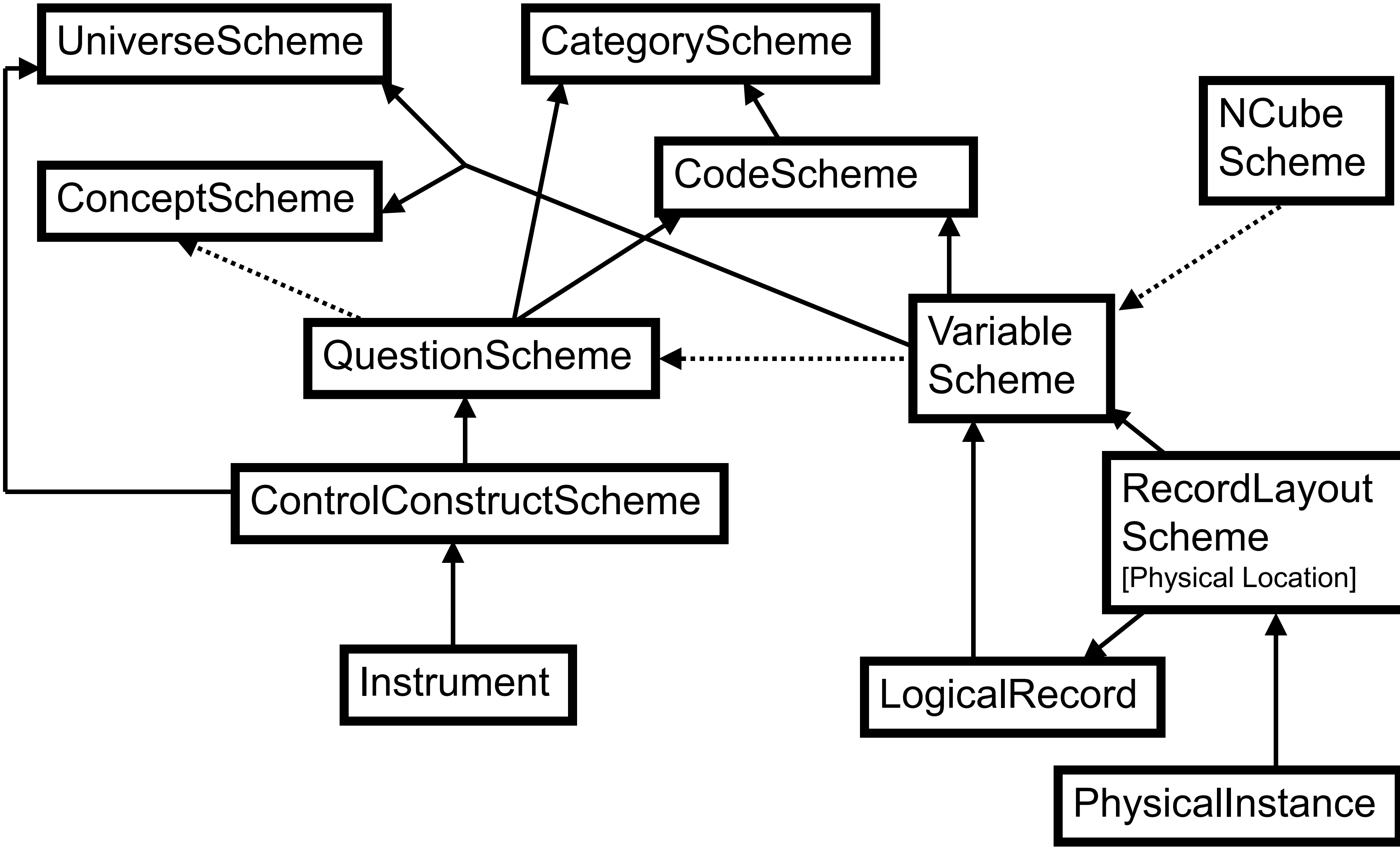
Modules

- Modules reflect the collection of metadata related to a specific stages of the lifecycle of a study or series of studies
- Contain reusable metadata (schemes) in-line or by reference
- Add information specific to the study
  - Identification, abstract, purpose, ownership, access
  - Lifecycle events, collection events, processing events (who, what, and when)
  - Descriptive content, summary statistics, internal relationships, external relationships

DDI Lifecycle Model and Related Modules



Building from Component Parts



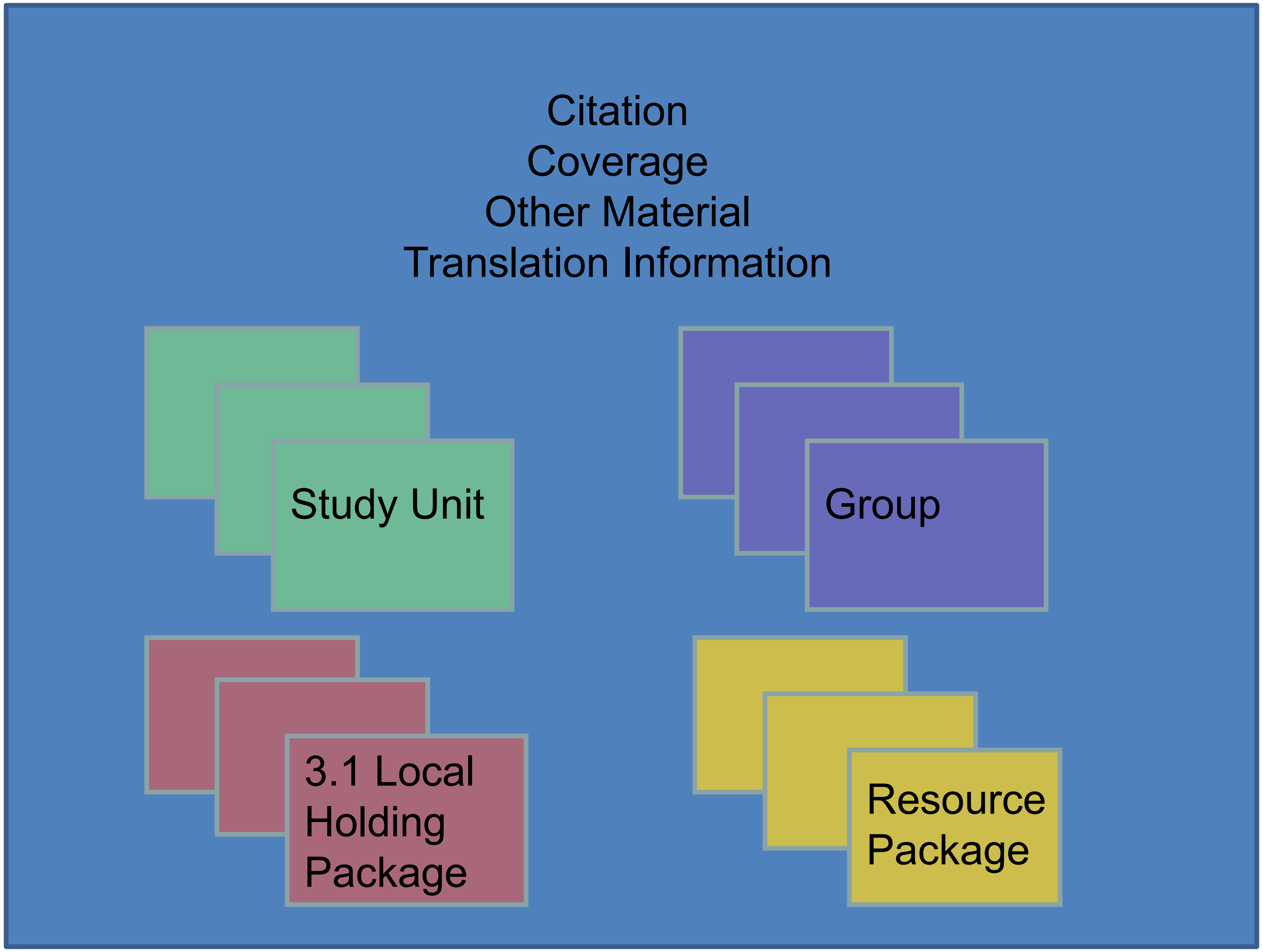
Module added content

- Citation
- Coverage
- Abstract / Purpose
- List of Resources Packages
- Methodology
- Data assessment
- Data relationship
- Collection content and relationships
- Access / Restrictions
- Archive / Library management details
- Comparison
- Other Material

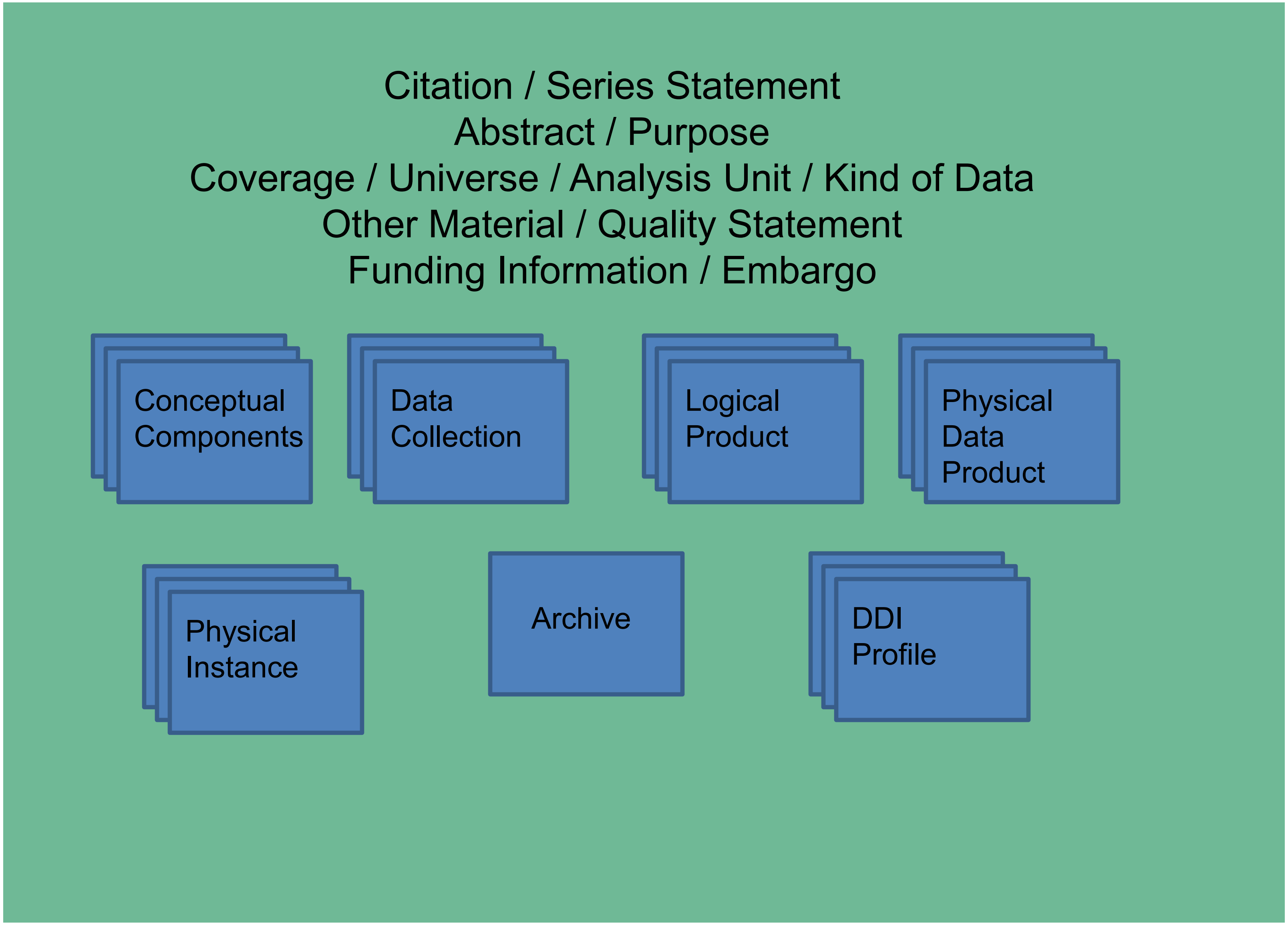
Packaging in XML for Publication

- There are five publication package modules in DDI and one exchange module
- DDI Instance is the top level packaging modules and serves as a consistent wrapper for the other four publication packages
  - Study Unit, Group, Resource Package, Local Holding Package
- Fragment Instance is the exchange module used to respond to a query (i.e., resolve an external reference

DDI Instance

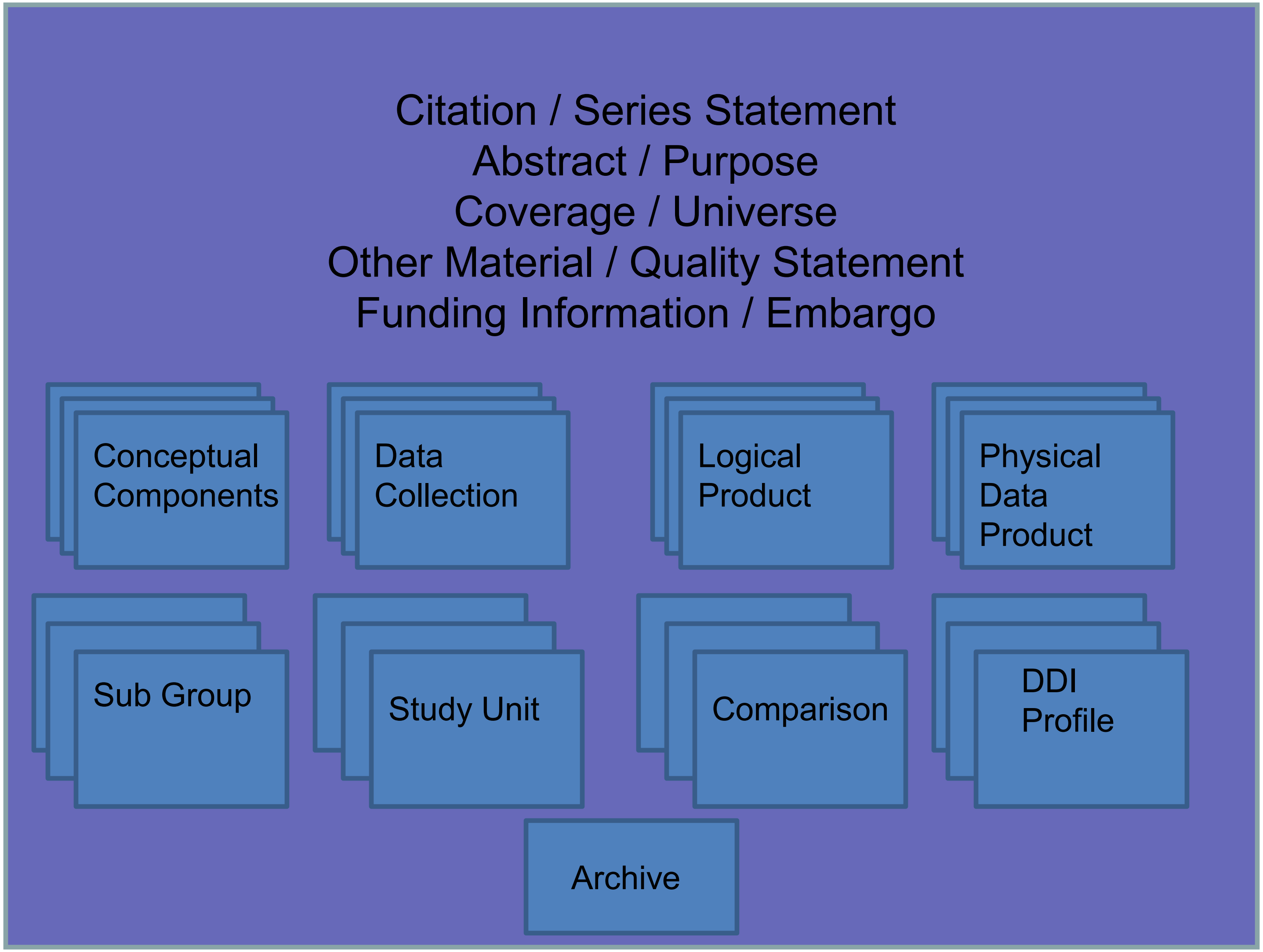


Study Unit



Citation / Series Statement
Abstract / Purpose
Coverage / Universe / Analysis Unit / Kind of Data
Other Material / Quality Statement
Funding Information / Embargo
Conceptual Components
Data Collection
Logical Product
Physical Data Product
Physical Instance
Archive
DDI Profile

Group


Citation / Series Statement
Abstract / Purpose
Coverage / Universe
Other Material / Quality Statement
Funding Information / Embargo
Conceptual Components
Data Collection
Logical Product
Physical Data Product
Sub Group
Study Unit
Comparison
DDI Profile
Archive

Resource Package

Citation / Series Statement
Abstract / Purpose
Coverage / Universe
Other Material / Quality Statement
Funding Information / Embargo

Any module
EXCEPT
Study Unit,
Group
Or
Local Holding
Package

Any Scheme:

Organization
Concept
Universe
Geographic Structure
Geographic Location
Question
Interviewer Instruction
Control Construct
Category
Code
Variable
NCube
Physical Structure
Record Layout
Quality Statement
Processing Instruction
Processing Event
Managed Representations

Local Holding Package (3.1 and later)

Depository Study Unit , Group Or Resource Package Reference:
[A reference to the stored version of the deposited study unit.]

Local Added Content:
[Study Unit, Group, or Resource Package content, plus a map relating new content to deposited content]

Advanced Topics

CAPTURING PROCESSES

Focusing on Processes

- Frequently DDI is first used to document a specific process
- A single process may be all that is under a users control
- Processes are often repeated and benefit most from metadata reuse
- Content generated by capturing a process can be used to “seed” the use of DDI both upstream and downstream from the initial area of use
- You have to start somewhere

Advanced Topics

CREATING A QUESTIONNAIRE

Questions

- DDI separates the questions themselves from the *use* of the questions in the questionnaire flow
  - This may seem odd, but there is a good reason for it!
- Questions can be reused!
- In DDI, the question itself is called a Question Item
  - Question Items are held in QuestionScheme
  - The QuestionScheme appears in the Data Collection module, inside of a Study Unit
- Complex question structures include MultipleQuestionItem in DDI 3.1, and QuestionGrid and QuestionBlock in DDI 3.2

Questionnaires

- Questions
  - Question Text
  - Response Domains
  - Relate to a Concept
- Statements
  - Pre- Post-question text
- Instructions
  - Routing information
  - Explanatory materials
- Question Flow
  - Conditional logic
  - Question acquires a Universe
- Instrument

Question Scheme Example

Look at the following questions:

B2. **Sempre nella settimana che va "DA LUNEDI'.... A DOMENICA...." Lei aveva comunque un lavoro che non ha svolto, ad esempio: per un periodo limitato di ridotta attività, per malattia, per vacanza, per cassa integrazione guadagni, etc.?**

- Sì 1
- No 2 *(passare a sezione E)*

B3. **Qual è il motivo principale per cui non ha lavorato in quella settimana?**

- Cassa Integrazione Guadagni (CIG ordinaria o straordinaria) 1 *(porre B4=1 e passare a B*
- Ridotta attività dell'impresa per motivi economici e/o tecnici (esclusa CIG) 2 *(passare a B4)*
- Controversia di lavoro 4 *(passare a B4)*
- Maltempo 5 *(passare a sezione C)*
- Malattia, problemi di salute personali, infortunio 6 *(passare a sezione C)*
- Ferie 7 *(passare a sezione C)*
- Festività nella settimana 8 *(passare a sezione C)*
- Orario variabile o flessibile (ad es. riposo compensativo) 9 *(passare a sezione C)*
- Part-time verticale 10 *(passare a B4)*

Question Item XML

```xml
<QuestionItem id="B2" version="1.0.0">
      <QuestionText>
      <Text>Sempre nella settimana che va  “DA LUNEDI’ ...
A DOMENICA..." Lei  aveva comunque un lavoro che non ha
svolto, ad esempio: per un periodo limitato di ridotta
attività, per malattia, per vacanza, per cassa
integrazione guadagni, etc.?</Text>
   </QuestionText>
   <CodeRespresentation blankIsMissingValue="true">
      <CodeSchemeReference>
         <ID>CodeSchemeYesNo</ID>
      </CodeSchemeReference>
   </CodeRespresentation>
   <ConceptReference>
      <ID>Concept_1</ID>
   </ConceptReference>
</QuestionItem>
```

Response Domains

- There are many possible types of response domains in the LFS:
  - Coded responses (as in the earlier example)
  - Date and Time responses
  - Numeric responses
  - Textual responses
  - Mixed responses (more than one type)
- Additionally, DDI can describe others as well:
  - Geographic responses
  - Categorical responses
  - Scales, Distribution, Ranking, and more (DDI 3.2)

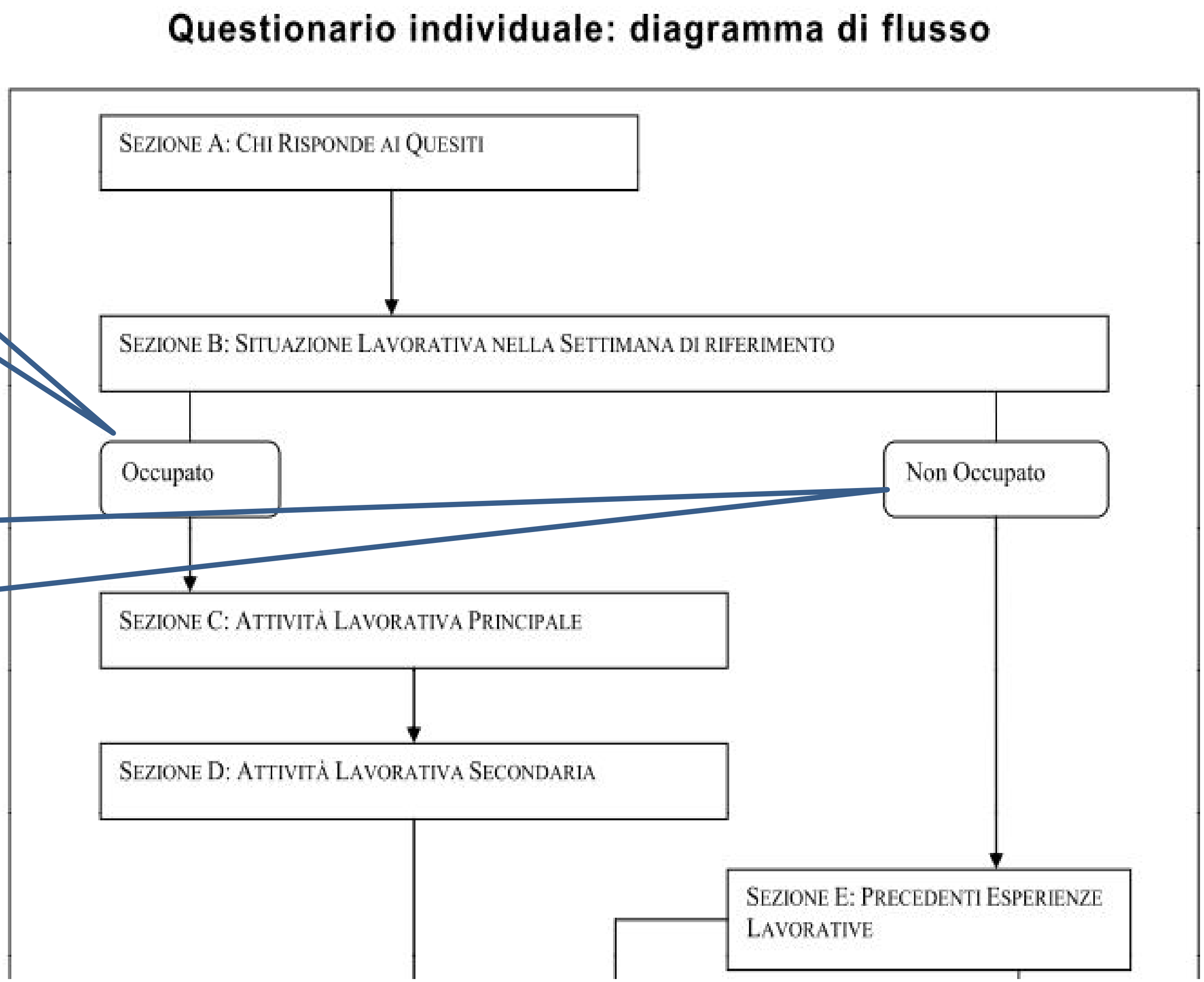
Interviewer Instructions

- "RISERVATO ISTAT" is an instruction to the interviewer that they are asked to fill in the question, not the respondent.
- This is an example of an Interviewer Instruction – text which tells the interviewer to do something, but which the respondent doesn't know about
- Interviewer Instructions are also used to express routing information, telling the interviewer or respondent which question to go to next
- Interviewer Instructions are found in Interviewer Instruction Schemes, in the Data Collection module
- DDI uses the term Instruction for the individual items in the scheme

Flow Logic

If you are employed, you answer sections C and D

If you are unemployed, you go directly to section E



Flow Logic Example

Here, if you answer “No” to question SG16, you must answer question SG 17, but if you answer “Si”, you go directly to question SG19 (with some qualifications about earlier answers):

SG16. Cittadinanza italiana

- Sì 1 *(se SG13=2 passare a SG18B; altrimenti passare a SG19_1)*
- No 2 *(passare a SG17)*

SG17. Cittadinanza straniera

RICODIFICATA NELLA VARIABILE CITSES

Flow Logic in DDI

- Flow logic is described in DDI using Control Constructs, found in Control Construct Schemes in the Data Collection module.
- Control Constructs can be of several different types:

Basic flow:

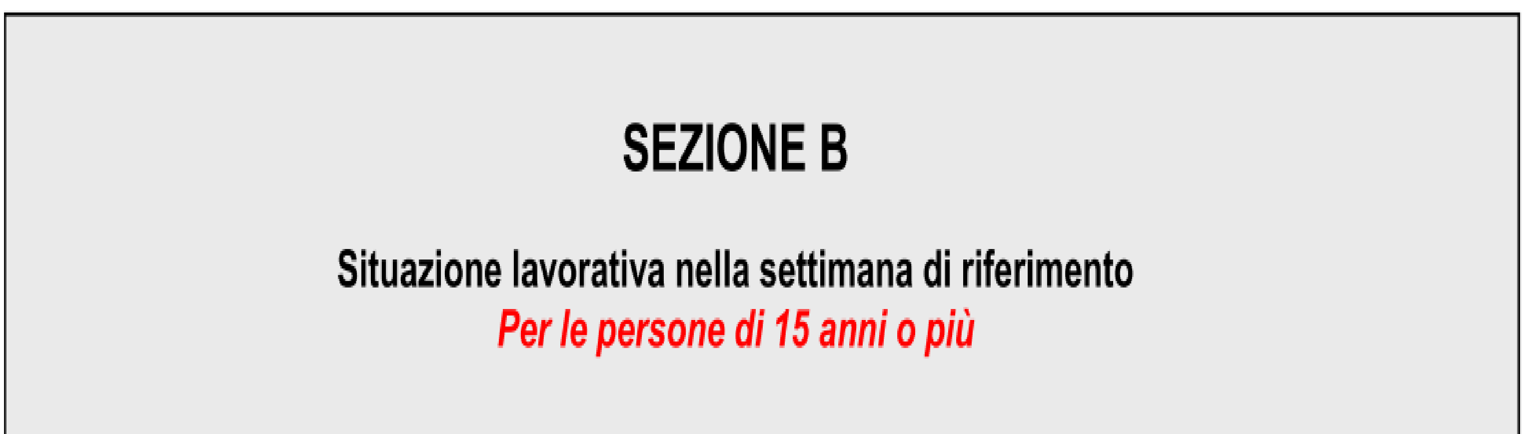
- Sequence
- Question Construct (the use of a Question Item)
- Statement Item
- Computation Item (for processing during administration)

Conditional logic:

- If Then Else
- Loop
- Repeat Until
- Repeat While

Sequences

- Every questionnaire starts with a top-level sequence
  - It may begin to branch conditionally very quickly
- Sequences reference other Control Constructs
- LFS starts with a reference to a Statement Item:



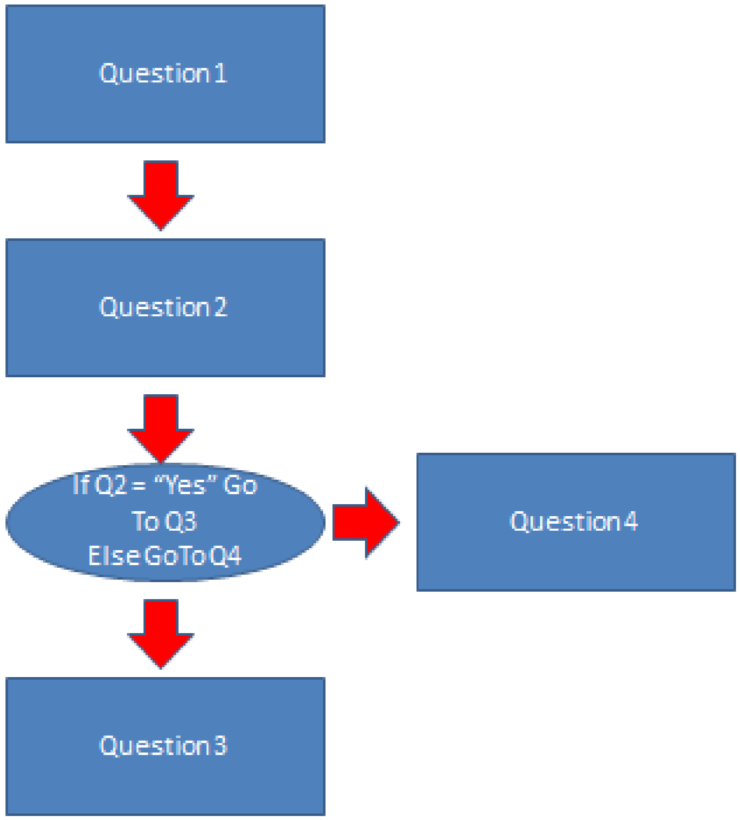
SEZIONE B

Situazione lavorativa nella settimana di riferimento

Per le persone di 15 anni o più

A Simple Example

Consider this simple example:



The Top-Level Sequence XML

```xml
<Sequence id="S1" version="1.0.0">
  <ControlConstructReference>
    <ID>Q1</ID>
  </ControlConstructReference>
  <ControlConstructReference>
    <ID>Q2</ID>
  </ControlConstructReference>
  <ControlConstructReference>
    <ID>ITE1</ID>
  </ControlConstructReference>
</Sequence>
```

Q1 is a Question Construct

Q2 is a Question Construct

ITE1 is an IF Then Else construct

Conditional Logic

- Conditional Control Constructs state the conditions which will determine the flow of the questionnaire
  - Conditions can be expressed using proprietary syntaxes, found in a Code element (ie, “if Q2 = 1”)
  - Users determine which syntax is needed
- In our simple flow example, if the answer to Question 2 was “Yes”, you go to Question 3 – otherwise, you go to Question 4.
  - This is an If Then Else construct

Conditional Logic XML

```xml
<IfThenElse id="ITE1" version="1.0.0">
  <IfCondition>
    <Code>Q2="Yes"</Code>
  </IfCondition>
  <ThenConstructReference>
    <ID>Q3</ID>
  </ThenConstructReference>
  <ElseConstructReference>
    <ID>Q4</ID>
  </ElseConstructReference>
</IfThenElse>
```

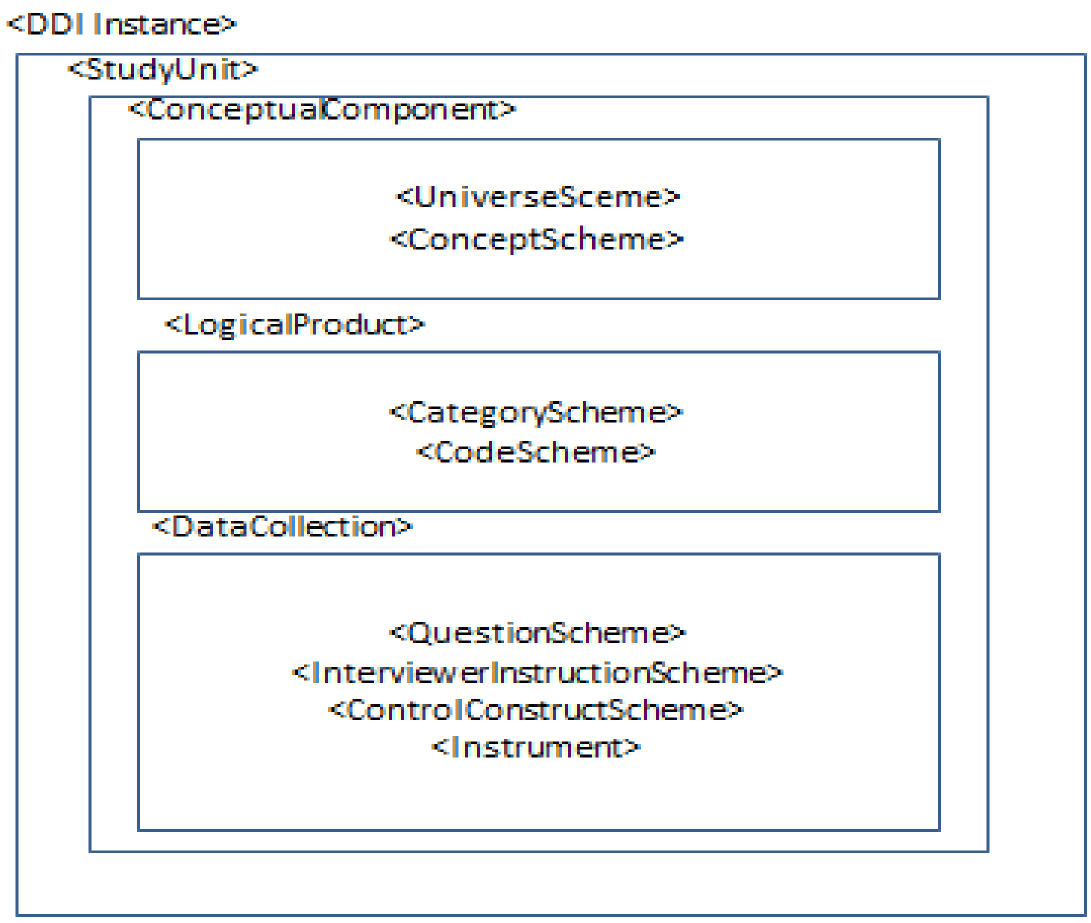
The Questionnaire

- In DDI, the questionnaire itself is represented by the Instrument
- This is found in the Data Collection module
- Each Instrument references the top-level Sequence in the Control Construct Scheme
- There will be a separate Instrument for each survey administration mode
  - But most of the other metadata will be reused

Instrument XML

```xml
<Instrument id="LFS" agency="ISTAT.IT" ↵
  version="3.0.2">
  <InstrumentName>
    <Name>Labour Force Survey Questionnaire for ↵
2012</Name>
  </InstrumentName>
  <Type>Print questionnaire form</Type>
  <ControlConstructReference>
    <Scheme>
      <ID>ControlConstructScheme1</ID>
    </Scheme>
    <ID>Seq1</ID>
  </ControlConstructReference>
</Instrument>
```

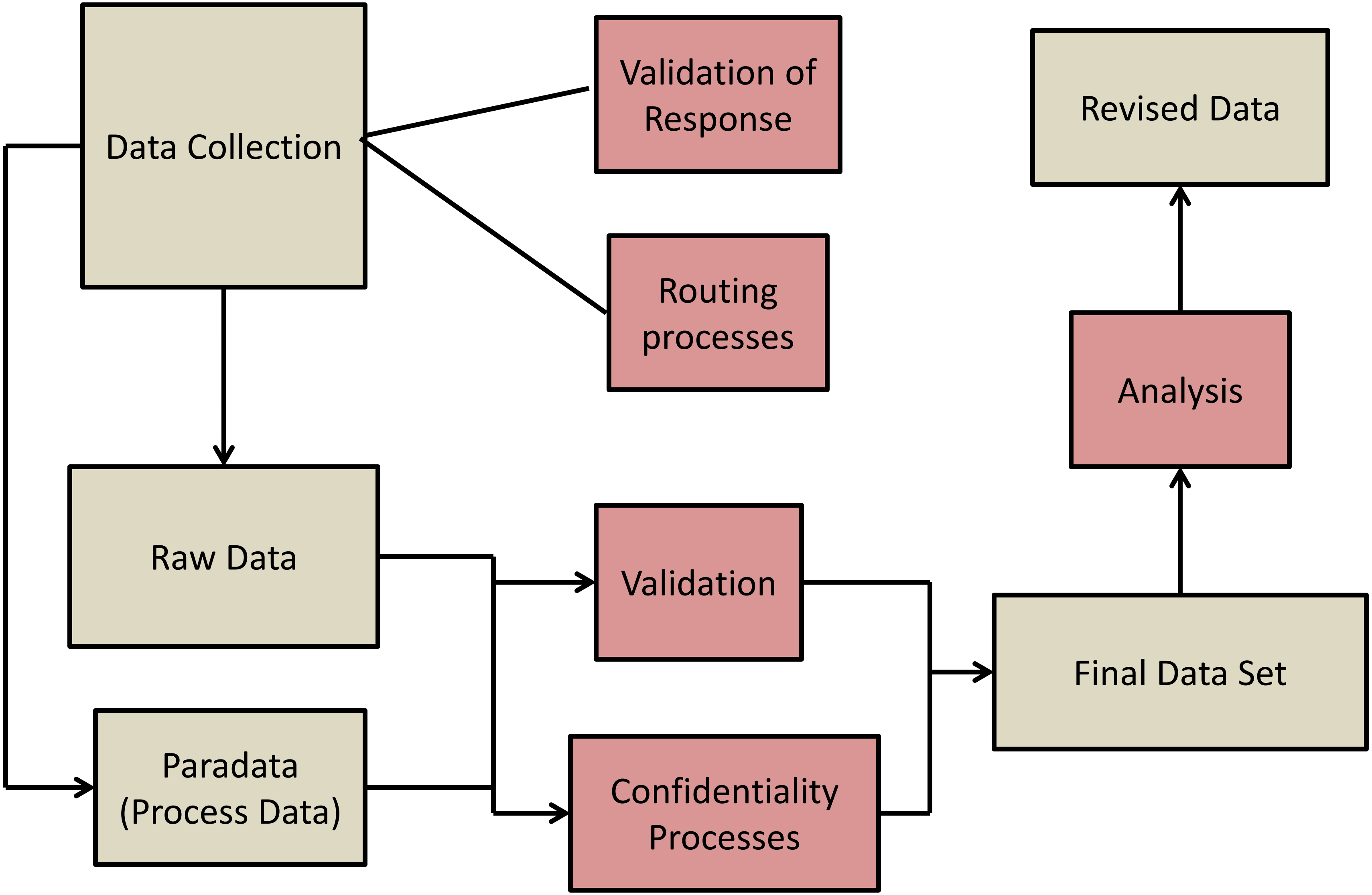
Bringing the Pieces Together



Functional Task 5

DOCUMENT DATA PROCESSING AND LIFE CYCLE EVENTS

Where does processing take place?



How is this handled in DDI?

- Processing Events
- Processing Instructions
  - Generation Instruction
  - General Instruction
- General descriptions
  - E.g. Collection Event/Action To Minimize Loss
- Life Cycle Event
  - Provides Date, Actor, Event description, Relationship (links to a detailed event description, Other Material, Generation Instruction, etc.)

Processing Event

- Processing Events bundle together information on a number of different events. These are bundled to reflect those activities which occur together:
  - Control Operation (type, description)
  - Cleaning Operation (type, description)
  - Weighting (description, standard weight factor)
  - Data Appraisal Information (Response, error, etc.)
  - Processing Instruction Reference
  - Quality Statement Reference
- DDI currently does not have a structured process model so Processing Event is descriptive only

Processing Instructions

- Processing Instructions are maintained in a scheme and are referenced from their point of use
  - Representation Map (comparative)
  - Processing Event
  - Variable Representation
  - Coverage / Restriction Process
  - Missing Values Representation

Command Code

- Note that both General and Generation Instructions contain Command Code
- Command Code is also available in:
  - Universe Generation Code
  - Computation Item
  - Condition for Continuation (of a loop or repeat)
  - Alternate Sequence
  - Conditions in routing with Control Constructs

```xml
<GenerationInstruction>
  <ID>GI_5to10</ID>
  ...
  <Description>Conversion of single year to 5     ↵
    year cohorts. Ages 0 to 20 plus. Includes     ↵
    inline command code and reference to          ↵
    external SPSS program file.</Description>
  <CommandCode>
    <ProgramLanguage>SPSS</ProgramLanguage>
    <CommandContent>IF (SOURCE LT 3) THEN 1;      ↵
      IF ((SOURCE GE 3) AND (SOURCE LT 4)) THEN 2; ↵
      IF (SOURCE EQ 5) THEN 3;</CommandContent>
  </CommandCode>
  <CommandFile>
    <ProgramLanguage>SPSS</ProgramLanguage>
    <URI>../Program/AgeConvert_1_5.sps</URI>
  </CommandFile>
</GenerationInstruction>
```

Life Cycle Event

- Life Cycle Event is a very general structure that was intended to capture significant events in the life cycle of the data and metadata
- It is currently the only place where we can capture the standard “process model” content of Object, Actor, and Event
- Located in the Archive module

```xml
<LifecycleEvent>
   <ID>LF_1</ID>
   ...
   <Label>Ingest of SIP to Archive</Label>
   <EventType>SIP.Validation</EventType>
   <Date>
      <SimpleDate>2012-07-06</SimpleDate>
   </Date>
   <AgencyOrganizationReference>
      <ID>IND_34</ID>
      ...
   </AgencyOrganizationReference>
   <Description>Validation of data against metadata provided in    ↵
      SIP</Description>
   <Relationship>
      <RelatedToReference>
         <ID>PROCEVENT_2</ID>
         ...
      </RelatedToReference>
      <RelationshipDescription>Process used to validate incoming   ↵
         data against provided metadata.</RelationshipDescription>
   </Relationship>
</LifecycleEvent>
```

Advanced Topics

DISCOVERY & COMPARISON

Citation and Coverage

- Five modules have citation and coverage sections
  - DDI Instance, Study Unit, Group, Resource Package and Physical Instance
  - The citation in Physical Instance is for the DATA FILE rather than the metadata
  - All citations can contain native Dublin Core elements
- Coverage is assumed to be expressed in its broadest form at the highest level (e.g., Group) and restricted in contained modules
  - Example: A Study Unit has a coverage of Resident Persons in the United States, 2000-2010
  - A Physical Instance may restrict this to Female Resident Population of the United States in 2008

Coverage

- Topical Coverage
  - Subject and Keyword
- Temporal Coverage
  - Reference Date with optional Subject
- Spatial Coverage
  - Description, Bounding Box, Country Code, Spatial Object, Geographic Structure Reference, Geographic Location Reference, Top and lowest level object, Geographic Structure Variable Reference, and Summary Data Reference

Discovery Metadata

Common Items

- Title
- Author
- Abstract
- Ownership
- Access requirements
- Link to object
- Topical Subject/Keyword
- Geographic coverage
- Temporal coverage
- Format information

Specialized Items

- Restricted subject list
- Information on the kind of data
- Geospatial coordinates
- Full variable/question description
- Collection detail
- Provenance chain

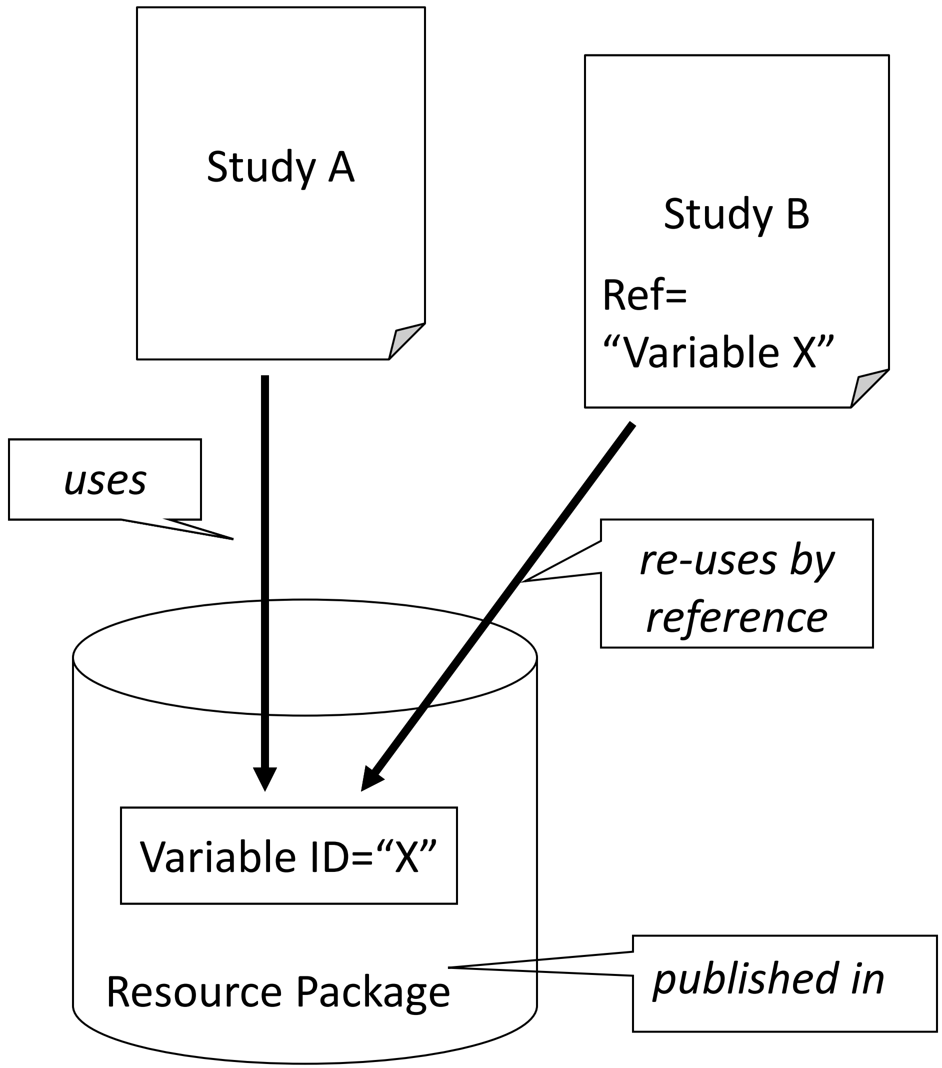
Common Metadata

- Accurate linking in the Linked Open Data world is facilitated by the use of common metadata resources
  - There are DDI based RDF vocabularies for discovering datasets and describing statistical classifications
- Fewer assumptions regarding the comparability of data objects from different sources
- Clarifies relationships of two objects within a larger structure (Geography, Organizations, Standard Classifications, etc.)
  - Immediately identifies if two data sets are using the same or different resources

Comparison in DDI

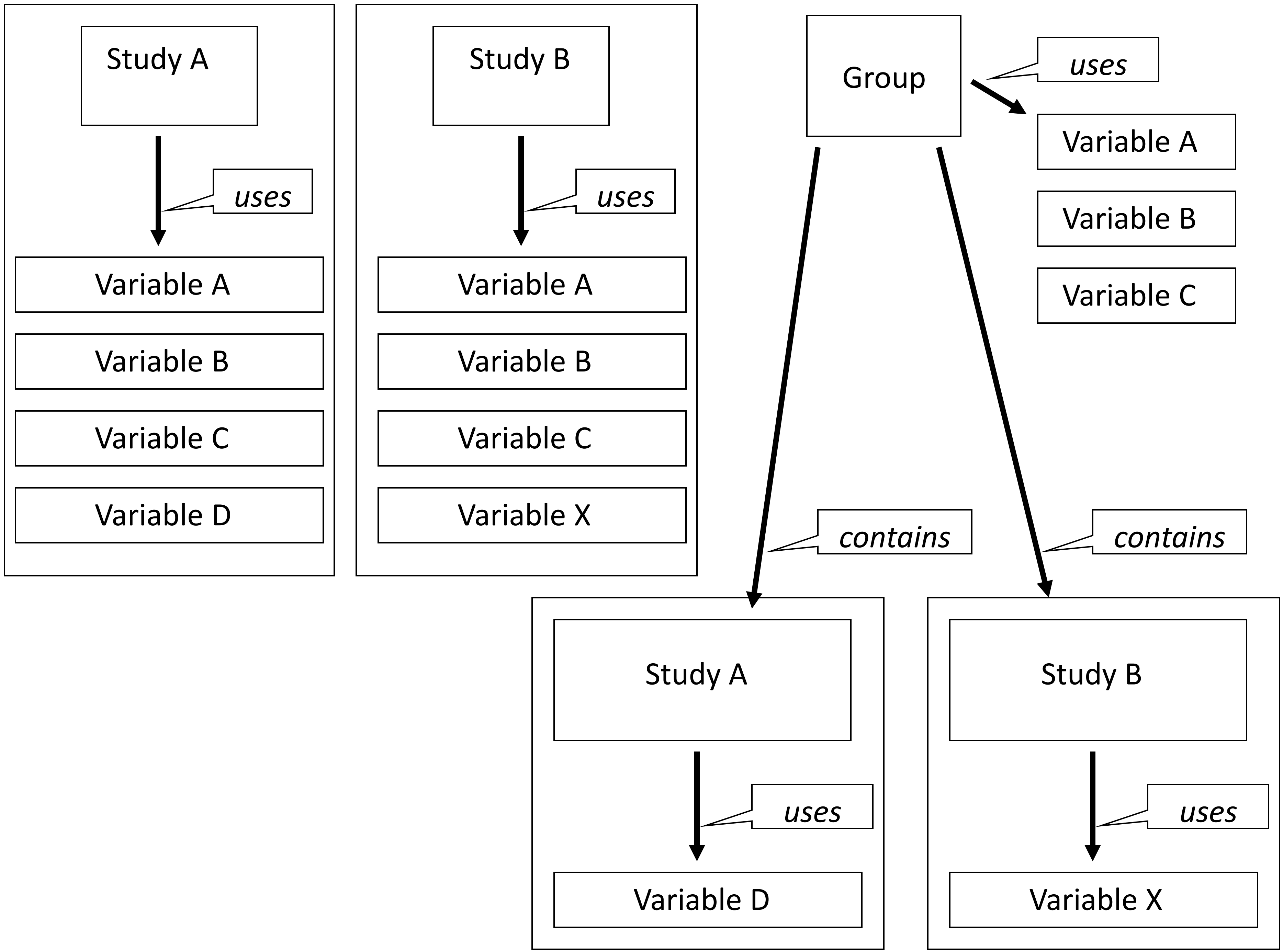
- Explicit comparison
  - Reuse of published common metadata
  - Grouping with inheritance
- Assertions of comparison
  - Comparison maps
  - Harmonization

Reusable Study-independent Information in Resource Package



Longitudinal (repeated) Studies

- The purpose of collecting the same data over time is to be able to make comparisons or evaluations of change
- These studies make extensive reuse of concepts, variables, questions, instruments, and even physical storage structures
- Rather than repeating content with each iteration, ***metadata is captured once and reused*** by each successive iteration. Reuse of an object implies comparability, changes are noted as they occur
- Statistical production is a very similar case


Study A
uses
Variable A
Variable B
Variable C
Variable D
Study B
uses
Variable A
Variable B
Variable C
Variable X
Group
uses
Variable A
Variable B
Variable C
contains
contains
Study A
uses
Variable D
Study B
uses
Variable X

Where is comparison useful?

- Describing differences between general or complex structures
  - Two concepts which are close but not exactly alike. Allows making assertions regarding the differences and similarities and lets the user decide if it is significant in terms of their research question.
- Explaining changes between versions of the metadata
  - Can provide mapping in both directions (old-to-new as well as new-to-old) to assist users in managing the effect of a version change on data collected before and after the change.
  - Mapping representation changes between versions
  - Explaining series breaks between cycles of data collection
- Harmonization
  - Mapping individual representation structures to a single unified structure (requires the creation of a unified structure which becomes the target structure of each map)

General Comparison

- Concept Map, Universe Map, and Category Map normally express general comparisons (i.e. the comparison can be captured in a single mapping or at most a set of maps describing each direction of the comparison).
- General comparisons identify the source and target objects being mapped and make assertions regarding the similarities and differences between them.
- Comparisons are always Source to Target.
  - If the direction of the comparison is immaterial, simply assign one to the source and one to the target.
  - For bi-directional comparisons create two maps, one in each direction
  - Bi-directional comparisons may or may not be meaningful

```xml
<UniverseMap>
   <ID>UniMap_1</ID>
   <MapName>US Population Clarification</MapName>
   <SourceSchemeReference>
      <ID>US_1</ID>
   </SourceSchemeReference>
   <TargetSchemeReference>
      <ID>US_2</ID>
   </TargetSchemeReference>
   <ItemMap>
      <ID>UniMap_1_1</ID>
      <SourceItem>
         <ID>U_1</ID>
      </SourceItem>
      <TargetItem>
         <ID>U_2</ID>
      </TargetItem>
      <Correspondence>
         <Commonality>Both cover Population in the United States, ↵
            Territories and Trust Lands</Commonality>
         <Difference>The Source Universe explicitly states that it is ↵
            the resident population. The Target Universe is unclear ↵
            on this point.</Difference>
      </Correspondence>
   </ItemMap>
</UniverseMap>
```

Complex Comparisons

- Comparing complex structures like Variables and Questions may require multiple maps
- For example we will look at a comparison of a number of questions all related to Age but using different question text and/or different response domains. This example is in DDI 3.1.
- DDI 3.2 allows comparison between different types of response domains (representations)
- DDI 3.2 allows listing the related maps used to compare specific aspects of the content.

Detail of question comparability

Comparison Map	Textual Content of Main Body		Category		Code Scheme	
	Same	*Similar*	*Same*	*Similar*	*Same*	*Different*
Question	**X**		**X**		**X**	
	X		**X**			**X**
	X			**X**	**X**	
	X			**X**		**X**
		X	**X**		**X**	
		X	**X**			**X**
		X		**X**	**X**	
		X		**X**		**X**

Harmonization

- Harmonization is the process of creating a common structure for coding so that data sets can be merged and analyzed as a single structure
- When harmonization results in a new data set with added “harmonized” variables the process is called “data fusion”
- When harmonization provides the information required to create a harmonized data set but does not take the step of creating the file is a process called “semantic integration”

Publishing Comparison

- Comparison can be published within a Group or Sub Group if it is related to a grouped set of studies (e.g. Waves in a longitudinal study)
- Comparison can be published as part of a Resource Package if it is related to differences between separate studies or to the description of a “virtual” harmonized set of data
- Systems, such as MISSY, capture the mapping between representations when a version change occurs. The mapping contains processing instructions which are “snippets” of statistical software programs used to create a harmonized variable.

Advanced Topics

CONCLUSION

Points to remember

- Few are starting from scratch
- Ongoing processes cannot stop
- You can only act in areas you control
- DDI is not an all or nothing structure

Check list

- Who do you need to interact with in your environment INPUTS and OUTPUTS?
- Where is your focus (may be different for different parts of the organization)?
- What do you control?
- What is your process flow - how far upstream is it practical to insert DDI like structures?

Advanced Topics

QUESTIONS?